Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

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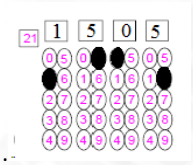
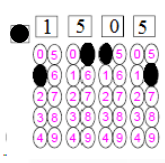
SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN** **Date: 24-01-2026**

Time: 09.00Am to 12.00Pm **GTM-18** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below

Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature: _____**

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MATHEMATICS : Total Syllabus

PHYSICS : Total Syllabus

CHEMISTRY : Total Syllabus

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MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(A)	Value of $\int_0^1 \frac{dx}{\sqrt{1-x^2}\left(1+\sqrt{1-x^2}\right)}$ is equal to	(I)	2
(B)	$f: R \longrightarrow R$ defined as $f(x)=x^5+e^{\frac{x}{2}}+e^{x^3}$ & $g(x)=f^{-1}(x)$, then value of $g'(2)$ is equal to	(II)	4
(C)	$\lim_{n\to\infty}\frac{1}{\sqrt{n}}\left(1+\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+\ldots\frac{1}{2\sqrt{n}}\right)$ equal to	(III)	12
(D)	If $f(x)=\int_{x-1}^{2x}\left(t^3-2t+4\right)dt$, then value of $f'(1)$ is equal to	(IV)	1

1) **(A)** - IV, **(B)** - I, **(C)** - II, **(D)** - III
2) **(A)** - I, **(B)** - II, **(C)** - IV, **(D)** - III
3) **(A)** - II, **(B)** - I, **(C)** - III, **(D)** - IV
4) **(A)** - III, **(B)** - IV, **(C)** - II, **(D)** – I

2. If $0<a<b<c<d$, then the quadratic equation $ax^2+\left[1-a(b+c)\right]x+abc-d=0$ has

1) real, distinct roots one of which lies between a and b

2) real, distinct roots with one of them between b and c

3) real, distinct roots one of them between c and d

4) roots which are not real

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3. If n is the number of terms in the expansion of $(1+x)^{101}\left(1+x^2-x\right)^{100}$ and if unit's place and ten's place digit in 9^n are λ and μ respectively, then which of the following is correct ?

1) $\lambda+\mu=9$ **2)** $\lambda-\mu=2$ **3)** $\mu-\lambda=1$ **4)** $\mu-\lambda=3$

4. Consider a tetrahedron ABCD whose vertex A is at origin. B,C lies on line $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z}{-1}$ such that vertices $A,B\,\&\,C$ form an equilateral triangle. Vertex D lies on $(x-1)^2+(y-5)^2+z^2\leqslant 0$. Then the volume of tetrahedron ABCD is equal to

1) $\frac{11}{32}\sqrt{3}$ **2)** $\frac{5}{16}\sqrt{3}$ **3)** $\frac{1}{18}\sqrt{15}$ **4)** $\frac{7}{36}\sqrt{15}$

5. Let ABC be a triangle with $AB=AC$ and circumcentre is at origin. Vertices A,B,C are represented by the complex number z_1,z_2,z_3 respectively then

STATEMENT-1 The complex number $\frac{(z_2+z_3)^2}{(z_1+z_2)(z_1+z_3)}$ is purely real

STATEMENT-2 $\angle DOF=\angle EOD$ where O is origin and D,E,F are the mid-points of the sides BC,CA and AB respectively.

1) Statement-1 is True, Statement-2 is True.
2) Statement-1 is False, Statement-2 is False.
3) Statement-1 is True, Statement-2 is False.
4) Statement-1 is False, Statement-2 is True.

6. Let α,β be roots of equation $x^2-x-3=0$. If $S_n=\frac{\alpha^{n+1}}{\alpha^{n+2}-\alpha^{n+1}-\alpha^n+\alpha^{n-1}}+\frac{\beta^{n+1}}{\beta^{n+2}-\beta^{n+1}-\beta^n+\beta^{n-1}}$, $n\in N$ then $|S_3|$ equals

1) $\frac{2}{9}$ **2)** $\frac{4}{9}$ **3)** $\frac{1}{9}$ **4)** $\frac{8}{9}$

7. For $0<x<\frac{\pi}{2}$, let $f_1(x)=\sum_{r=1}^{n}\sec\left(x+\frac{r\pi}{6}\right)\sec\left(x+(r-1)\frac{\pi}{6}\right)$. There exists $f_2(x),f_3(x)$ such that $f_1(x)-2f_2(x)=2\tan\left(x+\frac{n\pi}{6}\right)$ and $f_2(x)+f_3(x)=0$ in $\left(0,\frac{\pi}{2}\right)$ then $f_3(x)$ is

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1) Discontinuous and non-differentiable at $x=\frac{\pi}{4}$ and $x=\frac{\pi}{3}$

2) Neither continuous nor differentiable at $x=\frac{2\pi}{5}$

3) Continuous and differentiable in $\left(0,\frac{\pi}{2}\right)$

4) Continuous but not differentiable at $x=\frac{2\pi}{5}$

8. A circle C of radius 2 lies in the second quadrant and touches both the coordinate axes. Let r be the radius of a circle that has centre at the point (2,5) and intersects the circle C at exactly two points. If the set of all possible values of r is the interval (α,β), then $3\beta-2\alpha$ is equal to:

1) 15 **2)** 14 **3)** 12 **4)** 10

9. If the value of $(2020)^2+\frac{(2021)^2}{1!}+\frac{(2022)^2}{2!}+\frac{(2023)^2}{3!}+\ldots..$

is " $T\times e$ " then the sum of digits of T is

1)12 **2)** 55 **3)** 26 **4)** 101

10. Let $p(x)$ be 4th degree polynomial in x such that $\lim\limits_{x\to 0}\left(\frac{p(-x)}{2x^3}\right)^{\frac{1}{x}}=e^{-3}$ then the area bounded by $y=p(x)$ and x-axis in the second quadrant is

1) $\frac{1}{810}$ **2)** $\frac{1}{270}$ **3)** $\frac{1}{324}$ **4)** $\frac{1}{243}$

11. In a class there are four chairs C_1,C_2,C_3,C_4 and four students P_1,P_2,P_3,P_4. Initially, chair C_i is allotted to the student $P_i, i=1,2,3,4$. But on the examination day, these four students are randomly allotted four chairs. What will be the probability that P_2 gets previously allotted chair C_2 and NONE of the remaining students get the chair previously allotted to him/her is:

1) $\frac{1}{12}$ **2)** $\frac{2}{13}$ **3)** $\frac{5}{12}$ **4)** $\frac{11}{12}$

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12. Let f be a twice differentiable function satisfying $\frac{e^x f''(x) - e^x f'(x)}{e^{2x}} = 1$ and $f(0) = 0, f^1(0) = 1$ then $\int_0^1 \frac{f(x)}{x}\,dx$ is

1) e **2)** $e-1$ **3)** 1 **4)** 0

13. Let $A = \begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}$, and $X = \begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix}$. If P,Q are 2×2 matrices such that $P = AXB, Q = BX^T A$ and $\mathrm{Tr}\left((PQ)^{10}\right) = a^{10} + b^{10}$, where $a < b$, then $b - 2a = \left(\mathrm{Tr}(M)\right.$ denotes trace of matrix M and M^T denotes transpose of matrix M).

1) 1 **2)** 2 **3)** 3 **4)** 4

14. If $a_1, a_2, \ldots a_{10}$ are positive numbers in arithmetic progression such that $\frac{1}{a_1 a_2} + \frac{1}{a_2 a_3} + .. + \frac{1}{a_9 a_{10}} = \frac{9}{64}$ and $\frac{1}{a_1 a_{10}} + \frac{1}{a_2 a_9} + \ldots\ldots \frac{1}{a_{10} a_1} = \frac{1}{10}\left(\frac{1}{a_1} + \ldots. + \frac{1}{a_{10}}\right)$, then $\frac{a_1}{a_{10}} + \frac{a_{10}}{a_1}$ is

1) 5 **2)** $\frac{7}{2}$ **3)** $\frac{35}{16}$ **4)** $\frac{17}{4}$

15. Let P be image of $Q(7,-2,5)$ in $L: \frac{x-1}{2} = \frac{y+1}{3} = \frac{z}{4}$ and $R(5,\alpha,\beta)$ be a point on L. then $(area\ of\ \Delta PQR)^2$ is

1) 489 **2)** 957 **3)** 758 **4)** 924

16. Let $\frac{x}{2}\cos\theta + \frac{y}{1}\sin\theta = 1, \theta \in \left(\frac{\pi}{18}, \frac{\pi}{15}\right)$ intersect the ellipse $x^2 + 2y^2 = 6$ at P and Q , then angle between tangents at P and Q of the ellipse is -

1) $\frac{\pi}{16}$ **2)** $\frac{\pi}{6}$ **3)** $\frac{\pi}{8}$ **4)** $\frac{\pi}{2}$

17. CM and CN are the perpendiculars from the centre C of the ellipse $16x^2 + 25y^2 = 400$ to the tangent and normal at $P\left(\frac{5}{\sqrt{2}}, 2\sqrt{2}\right)$. The area of the rectangle CMPN is

1) $\frac{74}{25}$ **2)** $\frac{120}{41}$ **3)** $\frac{91}{25}$ **4)** $\frac{180}{41}$

18. Consider two lines $L_1 : \vec{r} = \left(\hat{i} - 2\hat{k}\right) + \lambda\left(2\hat{i} - \hat{j} + 2\hat{k}\right)$ and $L_2 : \frac{2x-1}{2} = \frac{y+1}{m}; z = 2$ where $\lambda, m \in R$

Sum of possible values of ' m ' for which shortest distance between L_1 and L_2 is $\sqrt{10}$ units is

1) 5 **2)**7 **3)** 6 **4)** 4

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19. For any increasing sequence of real numbers $a_1, a_2, a_3, \ldots$ let us denote A as $\prec\prec a_1, a_2, a_3, a_4, \ldots \succ\succ$, and a sequence ΔA is defined as $\prec\prec a_2 - a_1, a_3 - a_2, a_4 - a_3, \ldots\ldots \succ$. Suppose that all of the terms of the sequence $\Delta(\Delta A)$ are 1 and that $a_{19} = a_{92} = 0$. Then the sum of digit(s) of a_3 is M, then the value of $\frac{M}{2}$

1) 5 **2)** 22 **3)** 18 **4)** 12

20. Let α and β be the number of points of discontinuity and non-differentiability of $h(x)$ respectively, where $h(x) = \begin{cases} |x^2+5x-2| & , -8 \le x < 0 \\ \max(|x|, |x-2|), & 0 \le x < 2 \\ \min(|x-3|, 2), & 2 \le x < 6 \\ [2x]\operatorname{sgn}\left(x - \frac{15}{2}\right), & 6 \le x \le 8 \end{cases}$.

Then the value of $|\alpha - \beta|$ is where [.] greatest integer function and $\operatorname{sgn}(x)$ denotes signum

1) 7 **2)** 4 **3)** 5 **4)** 9

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. Number of 5 digit natural numbers whose sum of digits is 41 is____

22. The $\vec{A} = \hat{i} - 2\hat{j} + 3\hat{k}, \vec{B} = 2\hat{i} + \hat{j} - \hat{k}, \vec{C} = \hat{j} + \hat{k}$ If the vector $\vec{B} \times \vec{C}$ can be expressed as a linear combination **i.e.**, $\vec{B} \times \vec{C} = x\vec{A} + y\vec{B} + z\vec{C}$ where x, y, z are scalars, then the value of $(100x + 10y + 8z)$ is

23. If number of solutions of $\sin x = -x^2 + 2x$, when $x \in [0, 2\pi]$ is λ and when $x \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ is μ, then $\frac{\lambda - \mu}{2} =$

24. Let $f(x) = \lim_{n\to\infty} \frac{x^{2n-1} + ax^3 + bx^2}{x^{2n} + 1}, n \in N$ is continuous for all $x \in R$. If point $A(-a, 3)$ and $B(b+1, -1)$ are points of local maximum and minimum of a cubic polynomial $y = g(x)$, then the value of $g(2)$ is

25. Let $f : R \to R$ be a thrice differentiable function such that $f(0) = 0, f(1) = 2, f(2) = -3, f(3) = 4$, $f(4) = -5, f(5) = 6$ then the minimum number of zeros of $(3ff'' + ff''')(x)$ is

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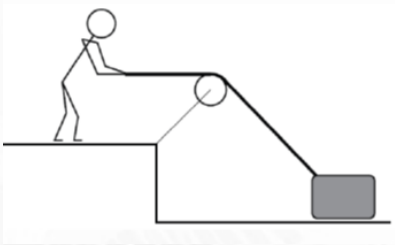
PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. As shown below, Arnav is using the rope through a fixed light pulley to move a box with constant speed v. The kinetic friction coefficient between the box and the ground is $\mu < 1$; assume that the fixed pulley is massless and there is no friction between the rope and the fixed pulley. Then, while the box is moving, which of the following statements is correct?



1) The magnitude of the force on the rope is constant.
2) The magnitude of friction between the ground and the box is decreasing.
3) The magnitude of the normal force of the ground on the box is increasing.
4) The pressure of the box on the ground is increasing.

27. Assertion (**A**) : A small metal ball is charged to a potential of +1 V. It is introduced into a large hollow metal sphere charged to a potential of +10,000 V and comes into contact with the inside surface of the sphere. The ball's charge passes to the sphere completely.
Reason (R) : Positive charge flows from a body at lower potential to another body at higher potential

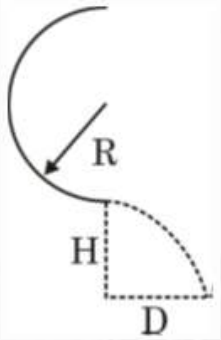
1) If both (**A**) and (R) are true and (R) is correct explanation of (**A**).
2) If both (**A**) and (R) are true but (R) is not the correct explanation of (**A**).
3) If (**A**) is true but (R) is false.
4) If both (**A**) and (R) are false.

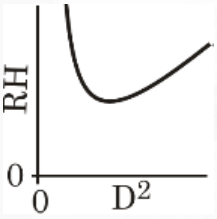
28. A spherical cloud of dust in space has a uniform density ρ and a radius R. The gravitational acceleration of free fall at the surface of the cloud due to the mass of the cloud is g. A process occurs (heat expansion) that causes the cloud to suddenly grow to a radius 2R, while maintaining a uniform density. The gravitational acceleration of free fall at a point R away from the center of the cloud due to the mass of the cloud is now :-

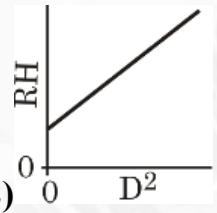
1) g/32 **2)** g/16 **3)** g/8 **4)** g/4

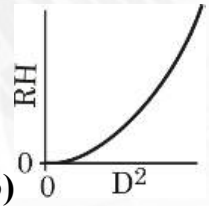
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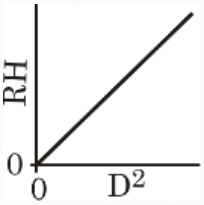
29. A smooth semicircular wire of radius R is oriented vertically. A small bead is released from rest at the top of the wire, it slides without friction under the influence of gravity to the bottom, where it then leaves the wire horizontally and falls a vertical distance H to the ground. The bead lands a horizontal distance D away from where it was released. Which of the following is a correct graph of RH against D?



1) 

2) 

3) 

4) 

30. Three-point charges +q, –2q and +q are placed at points (x = 0, y = a, z = 0), (x = 0, y = 0, z = 0) and (x = a, y = 0, z = 0) respectively. The resultant dipole moment can be best represented by qa $(\alpha \hat{i} + \beta \hat{j})$. Find α + β.

1) 2 **2)** 3 **3)** 4 **4)** 5

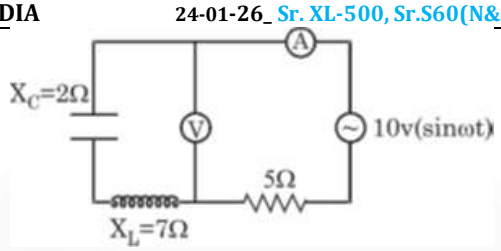
31. A paramagnetic sample shows a net magnetisation of 6 A/m when it is placed in an external magnetic field of 0.4 T at a temperature of 4K. When the sample is placed in an external magnetic field of 0.3 T at a temperature of 24K, then the magnetisation will be:

1) 4 A / m **2)** 0.75 A / m **3)** 2.25 A / m **4)** 1 A / m

32. In the figure shown hot wire voltmeter and hot wire ammeter are ideal. The reading of voltmeter is:

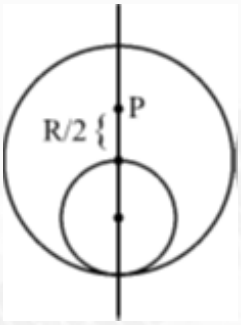
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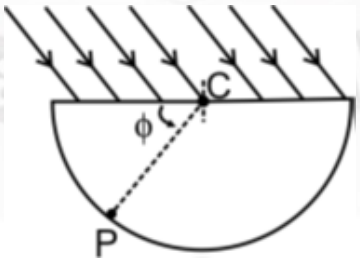
1) $5\sqrt{2}$ V **2)** 5 V **3)** 10 V **4)** None of these

33. A solid cylindrical wire of radius R has a current I flowing along its length. It has a long cylindrical cavity of radius R/2 as shown. Find the magnetic field at point P (in figure).



1) $\frac{\mu_0 I}{8\pi R}$ **2)** $\frac{\mu_0 I}{6\pi R}$ **3)** $\frac{\mu_0 I}{16\pi R}$ **4)** $\frac{\mu_0 I}{12\pi R}$

34. Parallel light rays are falling on the plane surface of semi cylindrical rod made of glass of refractive index 2.0 at an angle of incident of 60° in such a plane which is perpendicular to the axis of semi-cylinder as shown in figure. If φ is position where the rays are emerging out of cylindrical surface after refraction then φ lies between: [Given $\sin^{-1}(\frac{\sqrt{3}}{4}) = 25°$]



1) $85° < \phi < 145°$ **2)** $145° < \phi < 180°$ **3)** $75° < \phi < 105°$ **4)** $25° < \phi < 90°$

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35. A plane electromagnetic wave of wavelength λ has an intensity I. It is propagating along the positive Y-direction. The allowed expressions for the electric and magnetic fields are given by:

1) $\vec{E}=\sqrt{\frac{2I}{\varepsilon_0 c}}\cos[\frac{2\pi}{\lambda}(y-ct)]\hat{k};\vec{B}=+\frac{1}{c}E\hat{i}$

2) $\vec{E}=\sqrt{\frac{I}{\varepsilon_0 c}}\cos[\frac{2\pi}{\lambda}(y-ct)]\hat{k};\vec{B}=+\frac{1}{c}E\hat{i}$

3) $\vec{E}=\sqrt{\frac{2I}{\varepsilon_0 c}}\cos[\frac{2\pi}{\lambda}(y-ct)]\hat{k};\vec{B}=-\frac{1}{c}E\hat{i}$

4) $\vec{E}=\sqrt{\frac{I}{\varepsilon_0 c}}\cos[\frac{2\pi}{\lambda}(y-ct)]\hat{i};\vec{B}=\frac{1}{c}E\hat{k}$

36. Liquid is completely filled in a vessel up to height H and a hole is made at a depth h below the free surface of the liquid. Given that area of container is A and area of orifice is A. Match the entries of List-1 with the entries in List-2.

	List-1		List-2
(P)	Velocity of efflux	(1)	Proportional to $\sqrt{Hh-h^2}$
(Q)	The range of liquid comming out of the orifice & filling on the ground	(2)	Proportional to A_0
(R)	If the hole is at the bottom of the tank, time t taken to empty the tank is	(3)	Proportional to $\sqrt{h}$
(S)	Horizontal thrust force on the system (Liquid + Container) due to the liquid flowing out of container	(4)	Proportional to $\sqrt{H}$

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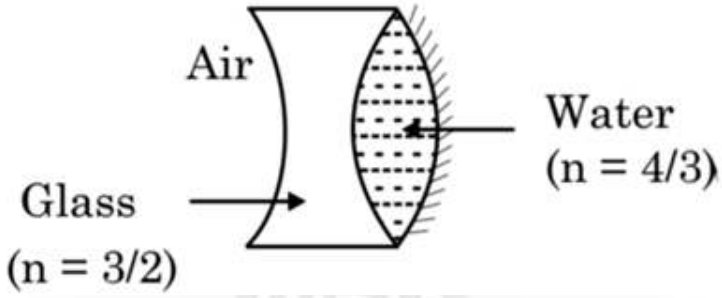
1) (P)→ (1), (Q)→ (3), (R)→ (4), (S)→ (2)
2) (P)→ (2), (Q)→ (1), (R)→ (4), (S)→ (3)
3) (P)→ (3), (Q)→ (1), (R)→ (4), (S)→ (2)
4) (P)→ (3), (Q)→ (2), (R)→ (4), (S)→ (1)

37. Two particles of mass m and 3m tied together with a very light string of linear mass density μ are dropped from a high rise tower as shown. A constant air friction force F acts on both of them. If the length of string is ℓ, the time taken by a pulse to go from bottom to top of string will be



1) $\ell\sqrt{\frac{\mu}{F}}$ **2)** $\ell\sqrt{\frac{2\mu}{F}}$ **3)** $\ell\sqrt{\frac{4\mu}{F}}$ **4)** $\ell\sqrt{\frac{\mu}{2F}}$

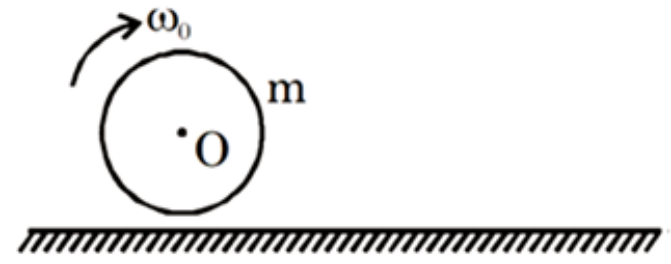
38. The radius of curvature of the left and right surface of the thin concave lens of refractive index 1.5 are 10 cm and 15 cm respectively. A spherical mirror of radius of curvature 15cm is placed in contact with the lens as shown in figure. The space between lens and mirror is filled with water of refractive index 4/3. This combination acts as



1) concave mirror of focal length 18 cm
2) concave mirror of focal length 36 cm
3) convex mirror of focal length 18 cm
4) convex mirror of focal length 36 cm

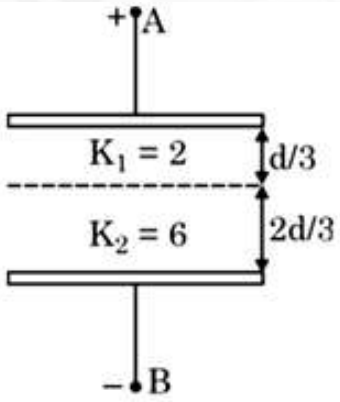
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39. A uniform solid cylinder of mass m and radius r is set in rotation about its axis with an angular velocity ω , then lowered with its lateral surface onto a horizontal plane and released. The coefficient of friction between the cylinder and plane is equal to μ. The time after which the cylinder starts rolling without slipping is



1) $\frac{\omega_0 R}{\mu g}$ **2)** $\frac{2\omega_0 R}{3\mu g}$ **3)** $\frac{\omega_0 R}{3\mu g}$ **4)** $\frac{3\omega_0 R}{4\mu g}$

40. A plate of capacitor having plate area A and gap between them is d. Two dielectrics of dielectrics constant K_1 & K_2 are filled between plates as shown in fig. If σ is charge density at positive plate of capacitor, then charge density at interface of dielectric is



1) $\sigma/3$ **2)** $\sigma/6$ **3)** $-\sigma/6$ **4)** $-\sigma/3$

41. A spherical object of mass 1 kg & radius 1 m is falling vertically downward inside a viscous liquid in gravity ***free*** space. At a certain instant the velocity of sphere is 2 m/s. If coefficient of viscosity of liquid is $\frac{1}{12\pi} Ns/m^2$; then velocity of ball will be 1 m/s after a time

1) ln 2 s **2)** 3ln 2 s **3)** 2ln 2 s **4)** 2ln 3s

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42. **Statement-1**: In meter bridge experiment it is suggested to repeat the experiment until null point is achieved near mid-point.
Statement-2: Meter bridge experiment uses Wheatstone bridge principle
1)Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1.
2)Statement–1 is True, Statement–2 is True; Statement–2 is not a correct explanation for Statement–1.
3)Statement–1 is True, Statement–2 is False.
4)Statement–1 is False, Statement–2 is True.

43. In screw gauge when jaws are in contact, zero of circular scale is below the main line by 3 division. Assume circular scale has 100 division and pitch is 0.5 mm. When a ball is placed between jaws, MSR is 4.5 mm and CSR is 58 then diameter of ball is
1) 4.805 mm **2)** 4.775 mm **3)** 4.790 mm **4)** 4.558 mm

44. The α-decay of a certain nucleus X^{230} is followed by emission of two groups of α-particles – one group having α-particles of kinetic energy 6.8 MeV while the other having α-particles of kinetic energy 5.2 MeV. Following the emission of these particles the daughter nuclei are found in ground and excited states. The energy of gamma photon emitted is: -
1) 1.43 MeV **2)**1.53 MeV **3)**1.63 MeV **4)**1.73 MeV

45. One mole of Argon undergoes a process given by $PV^{3\backslash 2}$ = constant. If heat obtained by gas is Q and molar specific heat in the process in C, then which of the following is correct if temperature of gas changes by –26K? (R → gas constant)
1) C = 0.5 R, Q = 13R **2)** C = –0.5 R, Q = 15R
3)C = –0.5 R, Q = 13R **4)** C = 0, Q = 13R

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**
Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

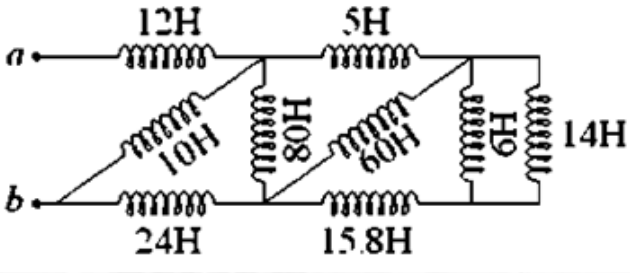
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46. A man with mass m jumps off of a high bridge with a bungee cord attached to his ankles. The man falls through a maximum distance H at which point the bungee cord brings him to a momentary rest before he bounces back up. The bungee cord is perfectly elastic, obeying Hooke's force law with a spring constant k, and stretches from an original length of L to a maximum length L = L + h. The maximum tension in the Bungee cord is four times the weight of the man. If the spring constant k= nmg/H, then find n.

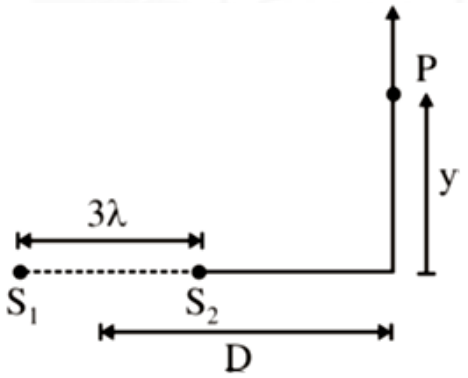
47. The work function of a metal is 3 eV and wavelength of subjected radiation to the metal is 275 nm. Then the de-Broglie's wavelength (in Å) of fastest photoelectrons emitted by the metal is nearly (Given : hc = 1240 eV – nm)

48. The equivalent inductance (in H) of the circuit shown below is



49. Find percentage error in measurement of moment of inertia of an annular disc. If measured value of inner radius (3.0 ± 0.1) cm, outer radius is (4.0 ± 0.1) cm and mass is (50 ± 0.1) g. (Round off to nearest integer)

50. Figure shows two coherent microwave sources S_1 and S_2 emitting waves of wavelength λ and separated by a distance 3λ. The minimum non-zero value of y for point P to be an intensity maximum is $\frac{\sqrt{x}}{2}D$, Then find value of x? (D >> λ)



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CHEMISTRY

Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

51. The reaction $N_2O_5(g) \rightleftarrows 2NO_2(g)+\frac{1}{2}O_2(g)$ is started with initial pressure of $N_2O_5(g)$ is equal to 600 torr. The fraction of $N_2O_5(g)$ decomposed when the total pressure of system is 960 torr is

1) 0.05 **2)** 0.1 **3)** 0.2 **4)** 0.4

52. Which one of the following complexes exhibits linkage isomerism

1) Cis $[CoCl_2(en)_2]^+$ **2)** Cis $[Co(en)_3]^{+3}$

3) Cis $[PtCl_2(NH_3)_2(py)_2]$ **4)** $[PtClBrINO_2]^{-2}$

53. Density of 2.05M solution of acetic acid is 1.02 gm/ml. Mole fraction of H_2O is

1) 0.96 **2)** 0.49 **3)** 0.89 **4)** 0.04

54. Consider the formation of SO_3 according to the reaction

$2SO_{2(g)}+O_{2(g)} \rightleftarrows 2SO_{3(g)};\ \Delta H=-198kJ$

Which of the following may change the value of equilibrium constant of the above reaction?

1) Doubling the volume of reaction vessel

2) Adding more of O_2 to the reaction vessel

3) Adding He gas to reaction vessel at constant pressure

4) Increasing the temperature

55. The vapour pressure of pure components 'A' and 'B' are 200 torr and 100 torr respectively. Assuming a solution of these components obey's Raoult's law. The mole fraction of component 'A' in vapour phase in equilibrium with a solution containing equimoles of 'A' and 'B' is

1) 0.33 **2)** 0.66 **3)** 0.80 **4)** 0.50

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56. Correct statements in the following are

a)In $Hg_{(l)} + Cl_{2_{(g)}} \rightarrow HgCl_{2(s)}$, the rate of disappearance of any of the reactants is same as the rate of appearance of the product

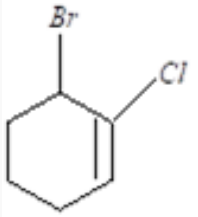
b)In $2HI_{(g)} \rightarrow H_{2(g)} + I_{2(g)}$, the rate of consumption of HI is half of rate of formation of H_2 or I_2

c)In $\underset{(aq)}{5Br^-} + \underset{(aq)}{BrO_3^-} + \underset{(aq)}{6H^+} \rightarrow \underset{(aq)}{3Br_2} + 3H_2O(l)$, the rate of consumption of $Br^-(aq)$ is 1.66 times to the rate of a formation of Br_2

1) Only 'a' is correct **2)** a, b only correct

3) a, c only correct **4)** a, b, c are correct

57. The IUPAC name of the compound shown below is



1) 1- Bromo-2-chlorocyclohexene **2)** 6-Bromo-1-chlorocyclohexene

3) 3- Bromo-2-chlorocyclohexene **4)** 2- Bromo-1-chlorocyclohexene

58. The basic character of hydrides of the V groups elements decreases in the order

1) $NH_3 > PH_3 > AsH_3 > SbH_3$ **2)** $SbH_3 > AsH_3 > PH_3 > NH_3$

3) $SbH_3 > PH_3 > AsH_3 > NH_3$ **4)** $NH_3 > SbH_3 > PH_3 > AsH_3$

59. Which of the following shows maximum number of oxidation states?

1) Cr **2)** Fe **3)** Mn **4)** V

60. Which of the following statements are correct with respect to hybridisation of atomic orbitals?

i) The orbitals present in the anti-penultimate shell of the atoms only are involved in hybridization.

ii) Excitation of electron is not essential condition for hybridization.

iii) Half-filled orbitals can participate in hybridization

iv) Completely filled orbitals of valence shell of an atom only can take part in hybridization.

1) i, ii only **2)** iii, iv only **3)** ii, iii only **4)** ii, iii, iv only

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61. Which of the following will exhibit maximum ionic conductivity?

1) $K_4[Fe(CN)_6]$ **2)** $[Co(NH_3)_6]Cl_3$

3) $[Cu(NH_3)_4]Cl_2$ **4)** $[Ni(CO)_4]$

62. Which one of the following sequences for the given property is correct?

1) Density - $Pb > Sn > Ge > Si$

2) Covalent radius - $Pb > Sn > Si > Ge > C$

3) Melting point - $C > Si > Ge > Sn > Pb$

4) Acidic nature - $PbO_2 > SnO_2 > GeO_2 > SiO_2 > CO_2$

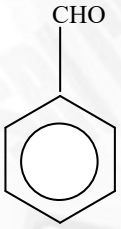
63. Choose the correct choice, which matches column-I with column-II.

	Mixture/compound (Column-I)		**Separation technique used (Column-II)**
P	Aniline + Water	i.	Sublimation
Q	Crude Oil	ii.	Distillation under reduced pressure
R	Camphor + NaCl	iii.	Steam distillation
S	$H_2O_2 + H_2O$	iv.	Fractional distillation

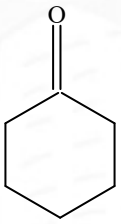
1) P-ii; Q-i; R-iii; S-iv **2)** P-iii; Q-iv; R-ii; S-i

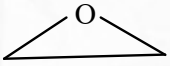
3) P-iii; Q-iv; R-i; S-ii **4)** P-iv; Q-iii; R-i; S-ii

64. Among the following compounds, the one which gives a tertiary alcohol on reaction with methyl magnesium bromide followed by hydrolysis is

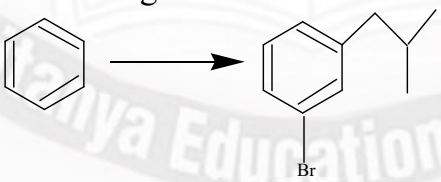
1) 

2) *HCHO*

3) 

4) 

65. The sequence of suitable reagents for the following conversion is



1) i. $(CH_3)_2CHCOCl/AlCl_3$; ii. $Br_2/FeBr_3$; iii. $NH_2.NH_2/KOH$;

2) i. $(CH_3)_2CH\text{-}CH_2Cl/AlCl_3$; ii. $Br_2/FeBr_3$

3) i. $(CH_3)_2CHCOCl/AlCl_3$; ii. $NH_2.NH_2/KOH$; iii. $Br_2/FeBr_3$

4) i. $Br_2/FeBr_3$; ii. $(CH_3)_2CH\text{-}CH_2Cl/AlCl_3$

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66. An organic compound having molecular formula C_3H_6O does not react with 2, 4 – dinitrophenyl hydrazine and does not react with Na metal. The compound is expected to be

1) CH_3-CH_2-CHO **2)** CH_3COCH_3

3) $CH_2=CH-CH_2-OH$ **4)** $CH_2=CH-O-CH_3$

67. Among the following amino acids, the one which has thiol group is

1) Cystine **2)** Cysteine **3)** Methionine **4)** Cytosine

68. Which of the following would have a permanent dipole moment?

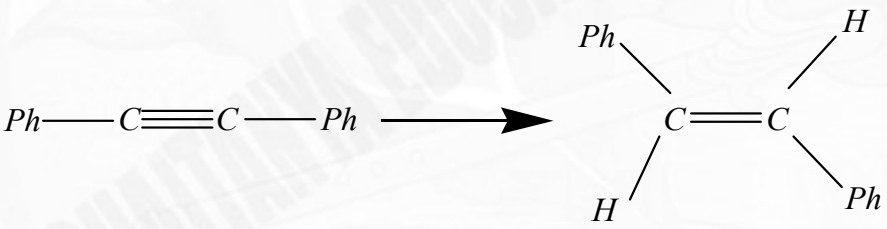
1) SiF_4 **2)** SF_4 **3)** XeF_4 **4)** BF_3

69. The nature of Cr^{2+} *and* Mn^{3+} respectively are

1) Oxidising and reducing **2)** Oxidising and oxidizing

3) Reducing and reducing **4)** Reducing and oxidizing

70. The reagent needed for converting



1) Cat. Hydrogenation **2)** H_2/Lindlar cat.

3) Li/NH_3 **4)** $LiAlH_4$

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. The bond order values of O_2, O_2^-, O_2^+ and O_2^{-2} and are a, b, c & d respectively. The value of a+b+c+d is

72. A molten salt of a metal is electrolyzed for 8 min with 20.0A current. 5.2g of metal is deposited if current efficiency is 50% only. The equivalent weight of metal is…..

(1F = 96000 Coulombs)

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73. The wavelength of electrons accelerated from rest through a potential difference of 40 kV is $x \times 10^{-12} m$. The value of x is _____ . (Nearest Integer)

Given: Mass of electron $= 9.1 \times 10^{-31} kg$

Charge on an electron $= 1.6 \times 10^{-19} C$

Planck's constant $= 6.63 \times 10^{-34} Js$

74. The oxidation state of 'Fe' in the product when Na_2S reacts with aqueous solution of $Na_2[Fe(CN)_5NO]$ is

75. The number of incorrect statement(s) is/are among the following is :

I. α-D- glucose and β-D- glucose are anomers.

II. α-D- glucose and β-D- glucose are enantiomers.

III. Cellulose is a straight chain polysaccharide made up of only β-D- glucose units.

IV. The penta acetate of glucose does not react with hydroxyl amine.

V. Amylopectin is a branched polymer of α- glucose.

VI. Cellulose is a linear polymer of β- glucose.

VII. Glycogen is the food reserve of plants

VIII. All proteins are polymers of α- amino acids.

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